

12.16

EE25BTECH11002 - Achat Parth Kalpesh

Question:

In a linear inverse problem, the coefficient matrix, $\mathbf{A} = \begin{pmatrix} 2.00 & 2.01 \\ 2.01 & 2.00 \end{pmatrix}$.

The eigenvalues of $\mathbf{A}$ are

- 1) $(4.01, -0.01)$ 2) $(-4.01, -0.01)$ 3) $(4.01, 0.01)$ 4) $(-4.01, 1.01)$

Solution

The characteristic equation for matrix $\mathbf{A}$ is given by

$$|\mathbf{A} - \lambda \mathbf{I}| = 0 \quad (4.1)$$

$$\begin{vmatrix} 2.00 - \lambda & 2.01 \\ 2.01 & 2.00 - \lambda \end{vmatrix} = 0 \quad (4.2)$$

$$(2.00 - \lambda)^2 - (2.01)^2 = 0 \quad (4.3)$$

$$((2.00 - \lambda) - 2.01)((2.00 - \lambda) + 2.01) = 0 \quad (4.4)$$

$$(-\lambda - 0.01)(-\lambda + 4.01) = 0 \quad (4.5)$$

$$\therefore \lambda_1 = 4.01, \quad \lambda_2 = -0.01 \quad (4.6)$$

$$(4.7)$$

Hence, the eigenvalues are

$$\lambda_1 = 4.01, \quad \lambda_2 = -0.01 \quad (4.8)$$